Customer Retention & Churn Analytics

End-to-end analysis with MySQL, Excel and Power BI

Presented by **Kavinprasath.R**

35,000
customers analysed

28.3%
churn rate

$67.9M
customer lifetime value

Project Overview and Objectives

Objectives

Measure churn

Churn rate, revenue and lifetime value lost

Find the drivers

Contract, plan, segment, payment and service features

Understand the reasons

Why customers leave and what each reason costs

Act early

Use the churn risk score to flag active at-risk customers

Tools and workflow

MySQL

Store, query, filter, join and rank the data

Microsoft Excel

Clean the data, pivot tables, charts, calculated metrics

Power BI

Two-page interactive dashboard for managers

METHOD

Dataset and Data Cleaning

35,000 customers

36 columns

5 regions

4 segments

6 churn reasons

Raw value	Cleaned to
we$$t, $outh$	West, South
ce ntr al, ba nk tran$fer	Central, Bank Transfer
@$tarTER, pl u@$	Starter, Plus
Device_p ro$tection	Device_Protection
Internet_Service blank (2,795)	Kept: no internet service
Duplicate Customer_IDs	None found

Customer_ID + 35 attributes

- Demographics (4)
- Account (4)
- Billing (7)
- Services (6)
- Behaviour and support (8)
- Outcome and value (6)

METHOD

SQL Analysis in MySQL

DDL and DML

CREATE, ALTER, INSERT, UPDATE and TRUNCATE to build the churn database

Filtering

LIKE, IN, BETWEEN and IS NULL to pull segments such as Basic on the Plus plan

Aggregation

COUNT, SUM, AVG, MIN and MAX on monthly charges

Grouping

GROUP BY region, plan and churn status; HAVING to filter groups

Window functions

RANK, DENSE_RANK, LAG, LEAD and running CLV by churn risk score

Joins

INNER, LEFT, RIGHT and FULL (UNION) to combine customers with services

Dashboard 1: Customer Loyalty & Churn Analysis



Power BI page: Customer Loyalty

Key insights

13.2K
Standard is the largest segment, then Basic with 12.6K

1.45M GB
Standard and Basic carry the most monthly usage

56%
of customers are aged 31 to 50 (27.8% and 28.2%)

961K
total tenure months across the customer base

Dashboard 2: Customer Churn & Retention



Power BI page: Dashboards (payment method slicer filters every visual)

Key insights

$2.09M
total monthly charges; $57.81M total charges

$67.88M
total customer lifetime value

~$2.3M
charges tied to each churn reason; the six are almost equal

~20%
each region's share of usage, so region does not separate customers

28.3% Churn Puts $14.4M of Lifetime Value at Risk

28.3%
Churn rate

9,898
Churned customers

$586K
Monthly revenue lost (28.0%)

$14.4M
Lifetime value lost (21.2%)

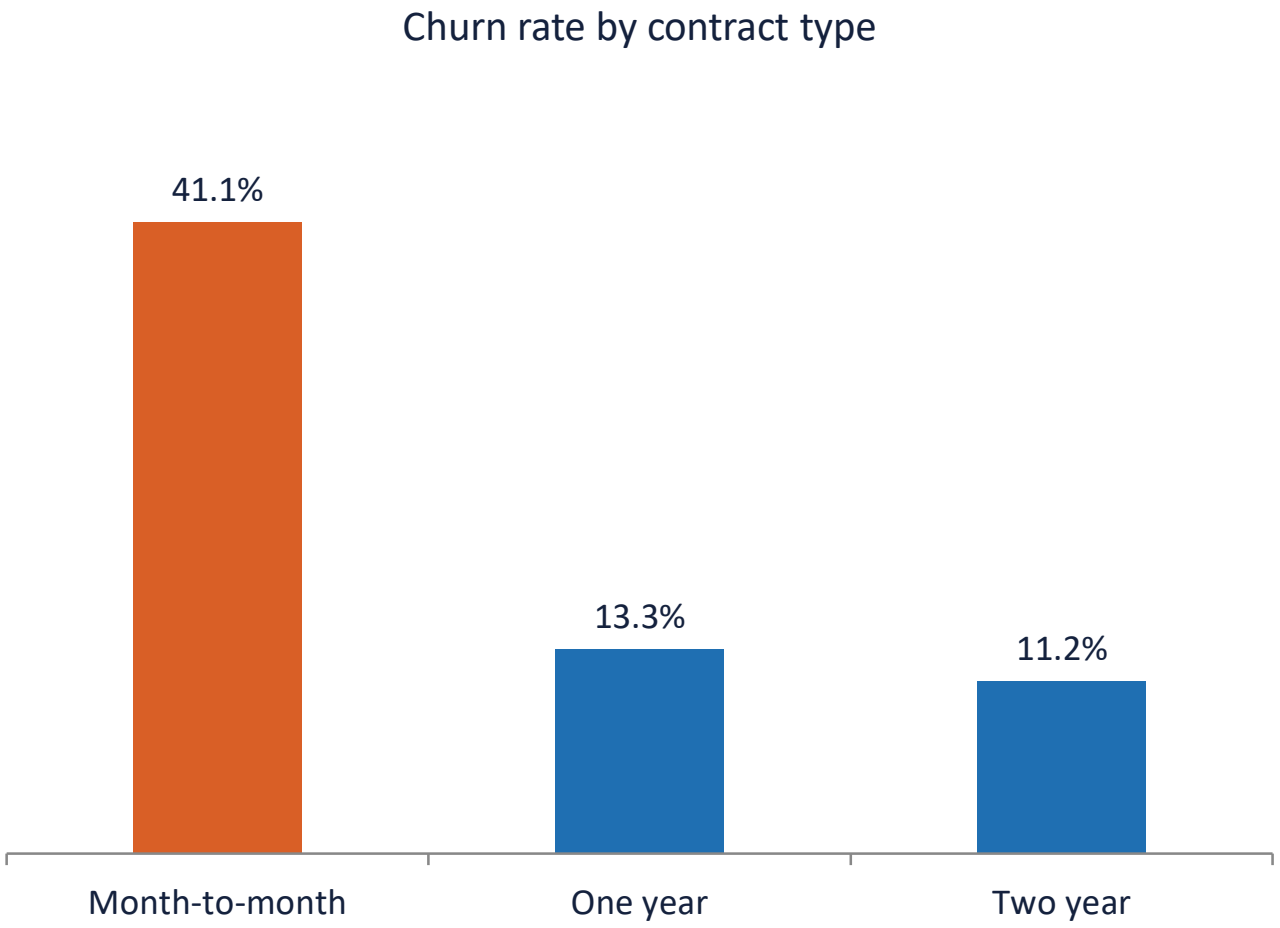
3.94 / 5
Average satisfaction

28.8
Average NPS score

Churned vs retained customers

Tenure 23.9 vs 28.9 months **Satisfaction 3.64 vs 4.06** **Complaints 1.42 vs 0.78 per year**

Month-to-Month Contracts Drive 80% of Churn



80%

of all churners are month-to-month (7,935 of 9,898)

55%

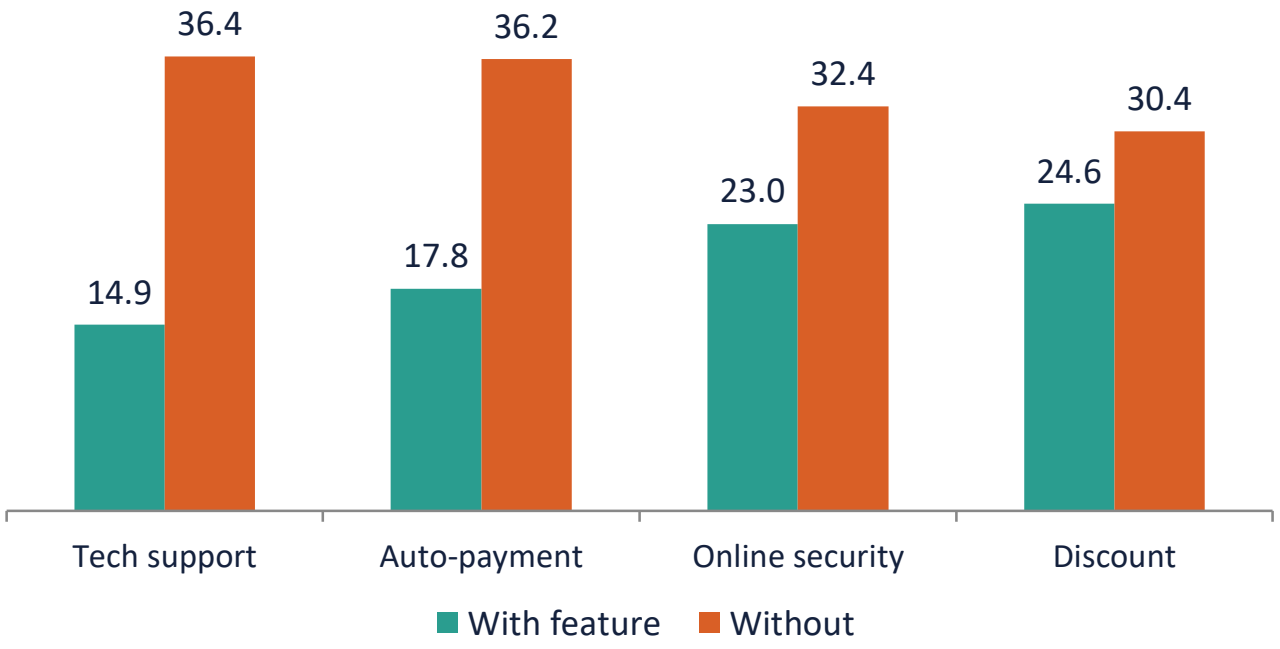
of customers are on month-to-month contracts

$11.6M

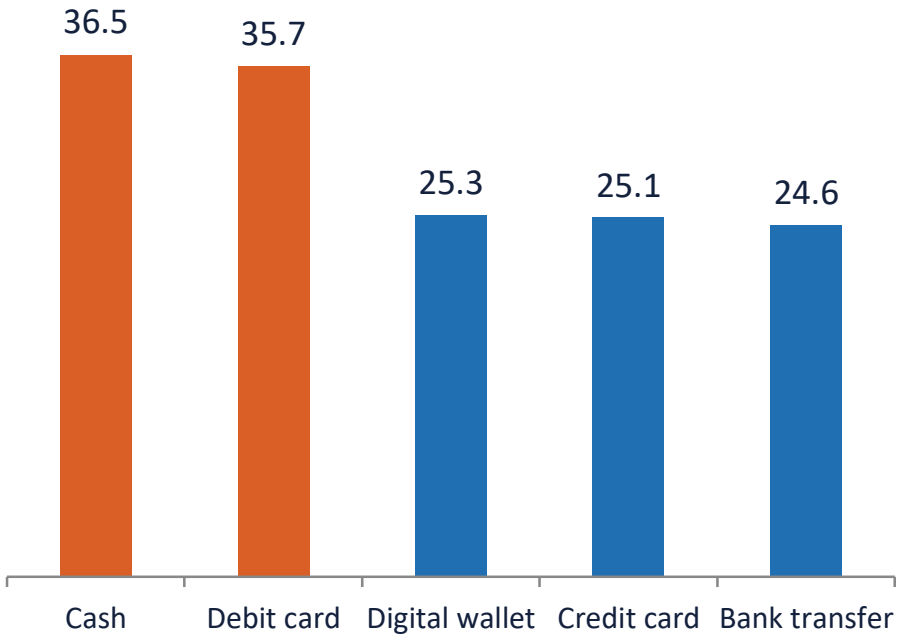
of churned lifetime value comes from month-to-month

Tech Support and Auto-Pay Cut Churn by More Than Half

Churn % with vs without feature



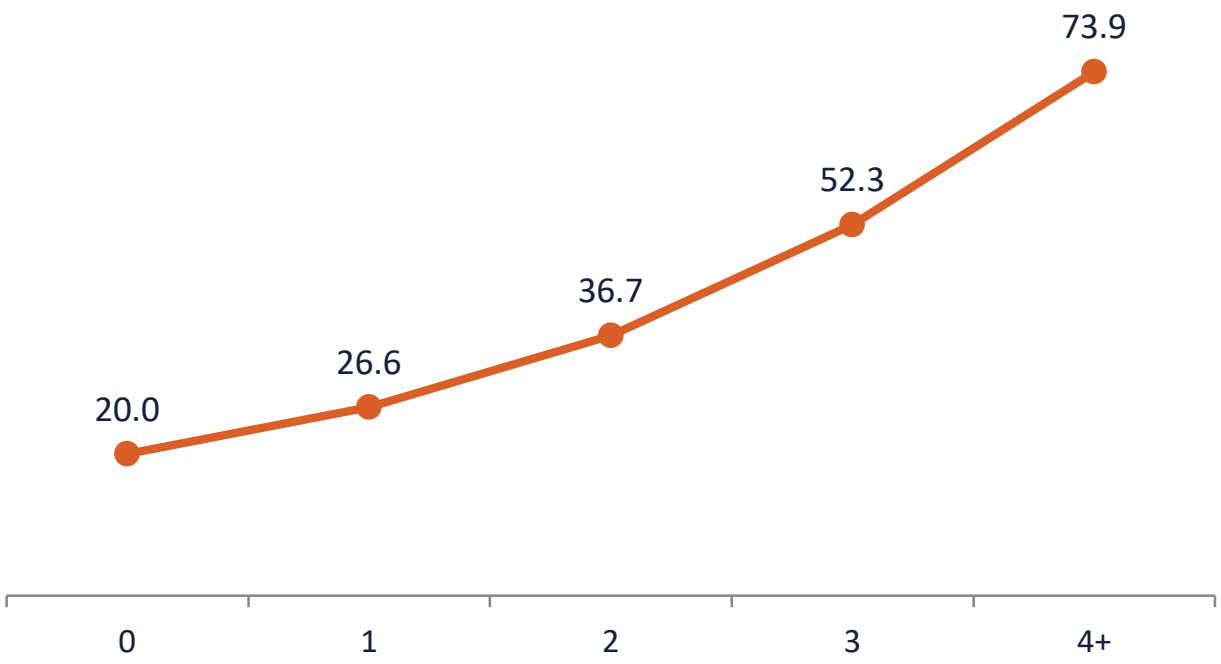
Churn % by payment method



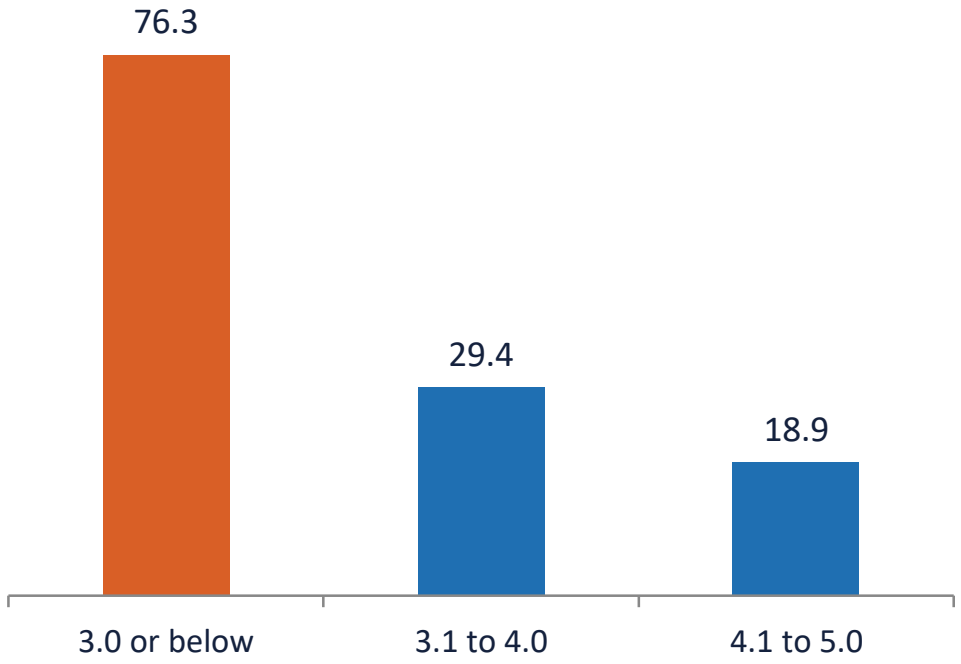
Customers with annual contract, auto-payment and tech support churn at just 2.4%.

Complaints Are the Clearest Warning Signal

Churn % by complaints last year

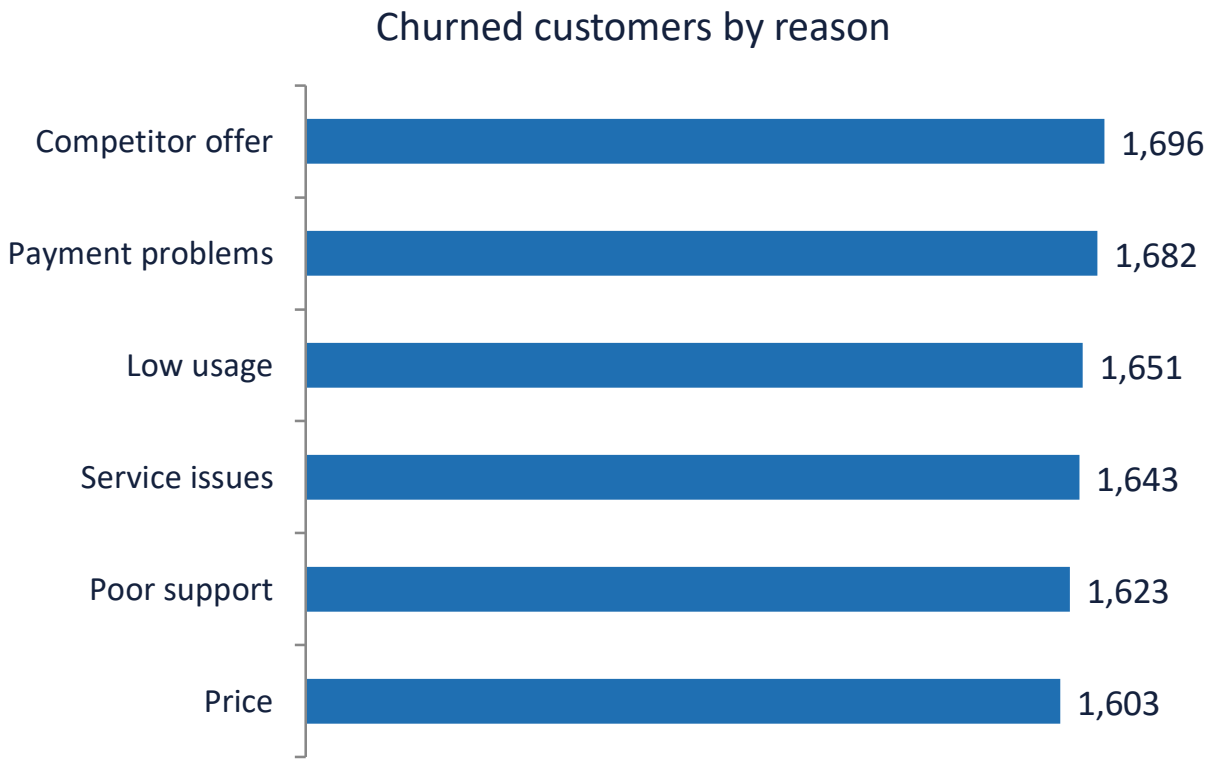


Churn % by satisfaction score



Support tickets: 17.0% churn at zero, 58.4% at five or more. Late payments: 20.9% rising to 49.5%.

Six Equal Reasons Mean Retention Needs Several Fixes

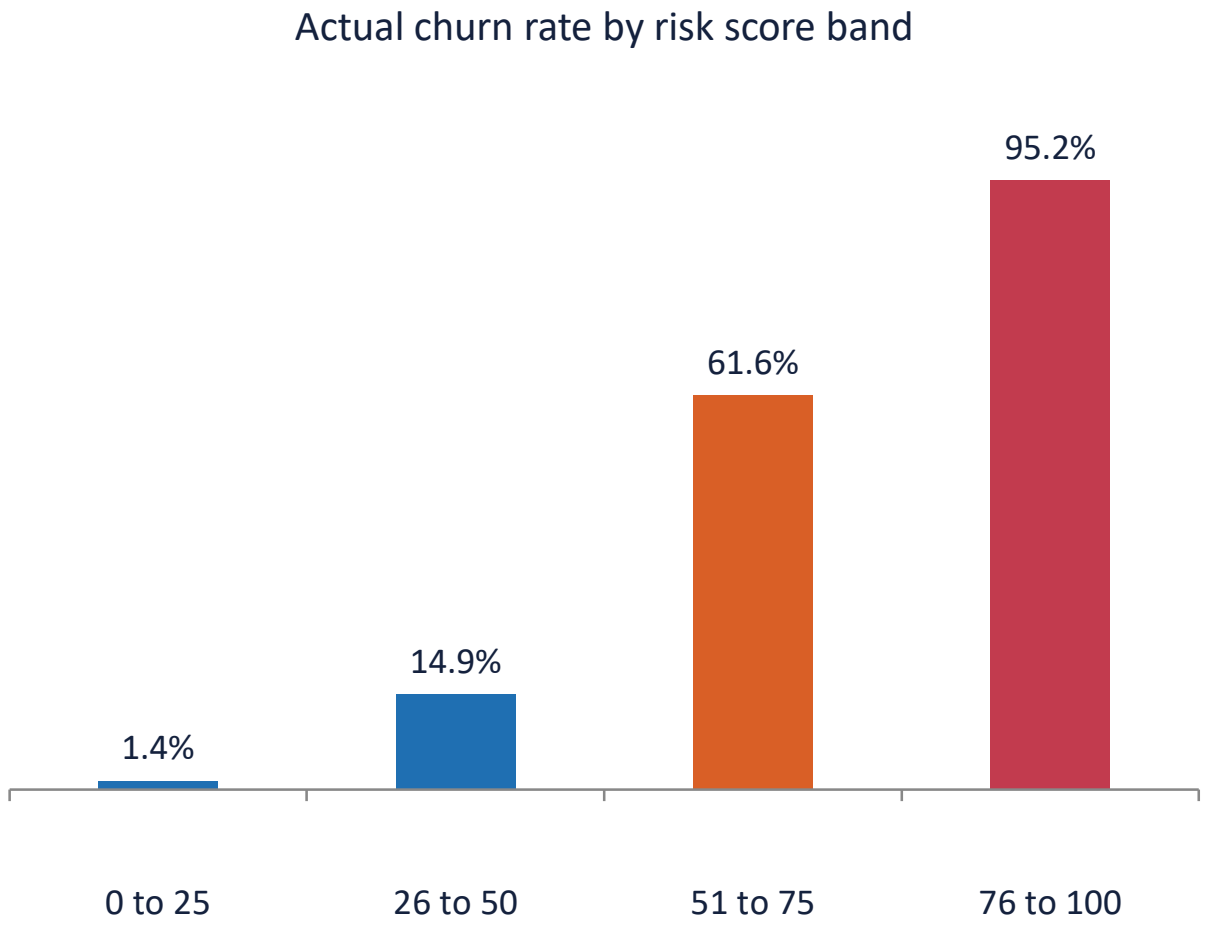


Segment	Customers	Churn	Avg CLV
Standard	13,181	29.0%	$1,670
Basic	12,617	28.5%	$1,435
Premium	7,095	28.3%	$2,559
Enterprise	2,107	22.4%	$4,557

Region, city tier, gender and age all sit within about 1 point of the 28.3% average. Target by behaviour, not demographics.

Each reason accounts for 16 to 17% of churners and about $2.3M to $2.5M of lifetime value.

3,511 Active Customers Can Still Be Saved



3,511

active customers with a risk score above 50

$209K monthly revenue | $6.1M lifetime value

Profile contrast

51.7% month-to-month, no auto-payment

2.4% annual contract, auto-payment and tech support

Recommendations

1. **Move month-to-month to annual plans**
   41.1% vs 13.3% churn

2. **Push auto-payment, card and wallet**
   36.2% vs 17.8% churn

3. **Bundle tech support**
   36.4% vs 14.9% churn

4. **Recover customers after complaints**
   Escalate at 2nd complaint or 3rd ticket

5. **Run an early-life programme**
   31.6% churn in the first 6 months

6. **Work the at-risk list**
   3,511 customers, $6.1M lifetime value

7. **Counter competitor and price exits**
   Targeted loyalty offers; six equal reasons

Target profile

Annual + auto-pay + tech support = 2.4% churn

Conclusion

- 28.3% of customers churn, putting $14.4M of lifetime value at risk
- Churn is driven by contract, payment and service experience, not by region or demographics
- Annual contract, auto-payment and tech support cut churn to just 2.4%
- 3,511 active high-risk customers can still be saved with early outreach

Next steps: load the full data into MySQL, build a predictive churn model, and auto-refresh the Power BI dashboard.

Thank you

Presented by

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